
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

Out-of-distribution (OOD) detection constitutes a central prerequisite for the robustness, calibration, and operational reliability of machine learning systems deployed beyond curated benchmark conditions. Recent work has increasingly investigated the incorporation of unlabeled data, thereby suggesting a promising route toward stronger OOD detection; nevertheless, the principled exploitation of unlabeled in-the-wild data remains difficult because such data typically commingles in-distribution (InD) observations with OOD samples. In the absence of a cleanly delineated OOD subset, the training of an effective OOD classifier becomes substantially more intricate. In this work, we introduce Medix, a median-centric framework that identifies potential outliers in unlabeled data through a median operation whose robustness to noise and atypical observations furnishes a stable estimate of central tendency. By coupling the outliers selected in this manner with labeled InD data, Medix trains a robust OOD classifier. Theoretically, we derive error bounds showing that Medix attains a low error rate, and empirically, we corroborate these guarantees by demonstrating that Medix consistently outperforms existing methods in open-world settings, thereby substantiating the practical relevance of the theoretical analysis.

1 Introduction

When machine learning models are deployed in real-world environments, their safety and reliability are frequently threatened by out-of-distribution (OOD) inputs whose categories were not represented during training and for which confident prediction is consequently undesirable. Despite this requirement, neural networks remain intrinsically susceptible to such inputs and, in practice, often lack mechanisms capable of recognizing and appropriately rejecting OOD examples (Nguyen et al., 2015).

The identification of OOD samples at inference time is indispensable but intrinsically challenging, since models trained without exposure to unknown distributions cannot, by construction, reliably discriminate those inputs from in-distribution (InD) data. To mitigate this limitation, recent approaches (Katz-Samuels et al., 2022a; Du et al., 2024a) have examined the use of additional “in-the-wild” data for improving OOD detection. In particular, Katz-Samuels et al. (2022a) proposed using unlabeled wild data to regularize model training while retaining the primary objective of classifying labeled InD examples. The practical appeal of such data lies in its ready availability: once a model is deployed in its operating environment, observations from that environment can be collected without explicit OOD annotation. This source of data can therefore help the model approximate the OOD distribution encountered at test time, yielding more robust OOD detection.

Nevertheless, the use of unlabeled wild data introduces substantial methodological complications because the resulting dataset is a heterogeneous mixture of InD and OOD observations. Without a distinct, uncontaminated collection of OOD samples, the construction of a reliable OOD detector is complicated by the fact that the detector observes only draws from the mixed distribution and is not

38 told whether any particular sample is InD or OOD. The setting consequently remains comparatively
 39 underexplored, leaving considerable room for algorithmic and theoretical development. Moreover,
 40 formal theoretical treatments are rare; to the best of our knowledge, Du et al. (2024a) is the only prior
 41 work that provides such a foundation for the “in-the-wild” setting.

42 Concurrently, recent work has shown that median-based procedures can be effective for data pruning
 43 (Acharya et al., 2024). Motivated by this observation, the present paper asks the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

44 To answer this question affirmatively, we introduce a median-centric perspective on OOD detection.
 45 More specifically, we formulate a median-based optimization framework and instantiate it through
 46 Medix, an algorithm that identifies OOD samples from wild data with a low error rate. We further
 47 provide a theoretical foundation that guarantees small error, and we validate this foundation experi-
 48 mentally by demonstrating the robustness and efficiency of the proposed algorithm. As one of the
 49 few works to supply theoretical guarantees for the unlabeled “in-the-wild” setting, our analysis shows
 50 that median-based filtering is robust for outlier detection in unlabeled mixtures by bounding the
 51 fraction of InD samples erroneously flagged as outliers. This fraction is governed by two effects: a
 52 contamination effect, which quantifies the influence of OOD points and remains manageable when
 53 the OOD proportion is below 50%; and a concentration effect, which uses the sub-Gaussian behavior
 54 of InD gradients to control deviations from their mean gradient. Together, these effects imply that,
 55 with high probability, the number of misclassified InD points remains small, even in adverse mixture
 56 configurations.

57 We benchmark our approach against two families of methods: (1) methods trained solely on InD
 58 data and (2) methods trained with both InD data and an auxiliary unlabeled dataset. On CIFAR-100
 59 (Krizhevsky et al., 2009), Medix substantially improves over the strong KNN+ baseline (Sun et al.,
 60 2022), reducing FPR95 by an average of 40.98%. Unlike approaches such as Outlier Exposure
 61 (Hendrycks et al., 2019), which require a clean auxiliary unlabeled dataset and thereby impose
 62 the strong distributional assumption that auxiliary data is fully separable from InD data, Medix
 63 achieves superior performance without relying on that assumption, affording greater operational
 64 flexibility. Relative to WOODS (Katz-Samuels et al., 2022a), Medix lowers average FPR95 by 1.32%
 65 on CIFAR-100 and by 2.60% on CIFAR-10.

66 Our key contributions are as follows:

- 67 **C1)** We propose Medix, a median-centric greedy procedure that first filters outliers from un-
 68 labeled wild data and subsequently trains an OOD detector using the identified outliers
 69 together with InD samples. The principal methodological contribution lies in the filtering
 70 stage.
- 71 **C2)** We establish theoretical guarantees for the robustness of median-based filtering when
 72 identifying both inliers and outliers in unlabeled mixtures. In particular, we prove that the
 73 misclassification rates for InD samples labeled as outliers and OOD samples retained as
 74 inliers are controlled by two effects: a contamination effect, which remains bounded while
 75 the OOD proportion is below 50%, and a concentration effect, which ensures stability in the
 76 gradient behavior of InD samples.
- 77 **C3)** We conduct an extensive evaluation of Medix across eleven InD–OOD pairs and compare
 78 its performance with 20 competitive baselines. The results show that Medix outperforms all
 79 baselines across the evaluated settings. We also demonstrate experimentally that Medix out-
 80 lier extraction attains a low error rate (e.g., only 12.5%; see Figure 2), thereby corroborating
 81 the theoretical findings.

82 2 Preliminaries and Problem Setup

83 This section first situates the OOD detection problem and then formalizes the data-generating setup,
 84 model classes, loss functions, and learning objective used throughout the paper.

85 **Labeled In-Distribution Data.** Let the input space $\mathcal{X}$ and label space $\mathcal{Y} = \{1, \dots, K\}$ specify the
 86 structure of the InD learning problem. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is obtained

by drawing n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. The marginal of $\mathbb{P}_{XY}$ over $\mathcal{X}$ is denoted by $\mathbb{P}_{\text{in}}$ and represents the underlying distribution of InD inputs. The dataset $\mathcal{S}_{\text{in}}$ is used to train the InD model.

Out-of-distribution detection. We consider the practical regime in which a model is trained on labeled InD data but is subsequently deployed in environments that may contain OOD inputs belonging to classes absent from the training label space, i.e., labels satisfying $y \notin \mathcal{Y}$. For such inputs, the model should abstain rather than issue an ordinary class prediction. Accordingly, the central inference-time task is to decide whether a given input is drawn from the InD distribution or from an OOD source.

Unlabeled wild data. A major obstacle in OOD detection is the scarcity of labeled OOD examples: because the potential OOD sample space can be extremely large, acquiring representative labeled instances is both costly and often infeasible. To address this difficulty, and to better approximate real deployment conditions as in Katz-Samuels et al. (2022a), we incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{x}_1, \dots, \tilde{x}_m\}$, into the learning framework. Wild data comprises a mixture of InD and OOD samples and can be collected during deployment of a model pre-trained on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a); Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize the marginal distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion and $\mathbb{P}_{\text{out}}$ denotes the OOD distribution on $\mathcal{X}$. The case $\pi = 0$ corresponds to the absence of OOD samples and is therefore a trivial instance of the problem.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameter $\phi \in \Phi$, where Φ is its parameter space. The output of f_ϕ is interpreted as a soft probability distribution over the $K = |\mathcal{Y}|$ InD classes, and the loss for labeled InD data is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$, where Θ denotes the corresponding parameter space. Its binary loss is written as $\ell_b(g_\theta(x), y_b)$ with $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, where $y_+ > 0$ represents the InD class and $y_- < 0$ represents the OOD class.

Learning objective. The learning framework jointly trains the OOD detector g_θ and the multi-class classifier f_ϕ by leveraging both the InD distribution $\mathbb{P}_{\text{in}}$ and the wild-data distribution $\mathbb{P}_{\text{wild}}$. At test time, performance is evaluated using the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ denotes the OOD test distribution.

3 Method: Median-Centric Framework for OOD Detection

This section introduces **Medix**, a median-centric learning paradigm for OOD detection that exploits unlabeled wild data while avoiding exclusive dependence on labeled InD examples. The framework is intended for open-world deployments, in which models routinely encounter inputs outside their training support, and consists of two interlocking stages: **1) Outlier Extraction**, a filtering procedure that isolates candidate OOD samples from unlabeled wild data (Section 3.1); and **2) Detector Training**, the training of a binary OOD detector using labeled InD data together with the candidate outliers identified in the first stage (Section 3.2). For the second stage, we follow the protocol introduced by Du et al. (2024a). As the subsequent sections show, this two-stage methodology both extracts OOD data effectively from unlabeled wild mixtures and provides a robust basis for deploying machine learning models in dynamic open-world environments.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, the framework employs an optimization-based procedure that operates on gradients of the model parameters. These gradients are induced by a classification model, f_ϕ , trained exclusively on the InD dataset $\mathcal{S}_{\text{in}}$. The resulting filtering methodology is specified formally below.

132 **Reference gradient estimation from InD data.** The first step estimates a reference gradient from
 133 the InD dataset $\mathcal{S}_{\text{in}}$. We obtain this reference by training a classifier f_ϕ on $\mathcal{S}_{\text{in}}$ via empirical risk
 134 minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

135 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training the classifier, we compute the mean gradient
 136 $\bar{\nabla}_{\text{in}}$ by averaging the loss gradients with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

137 In the proposed approach, $\bar{\nabla}_{\text{in}}$ functions as the reference gradient against which deviations associated
 138 with other data points are quantified.

139 **Motivation.** We posit that increasing the number of OOD samples in the wild dataset, $\mathcal{S}_{\text{wild}}$, should
 140 induce a larger deviation from the average InD gradient, $\bar{\nabla}_{\text{in}}$. To examine this hypothesis, we conduct
 141 an initial experiment with CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer
 142 et al., 2011) as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples that are
 143 disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$, which in turn is used to compute $\bar{\nabla}_{\text{in}}$. We then
 144 incrementally inject SVHN OOD samples into $\mathcal{S}_{\text{wild}}$ and monitor the L_2 -norm deviation between
 145 $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-dataset gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

146 The results in Figure 1 exhibit a clear monotonic increase in the L_2 -norm deviation, thereby supporting
 147 the hypothesis and motivating the filtering method introduced next. This monotonic behavior also
 148 informs the algorithmic stopping rule: the procedure terminates when the change in L_2 -norm deviation
 149 between consecutive iterations falls below the threshold ϵ , as detailed in the following section.

150 **Filtering potential outliers from unlabeled wild data.** Motivated by Figure 1, we cast the identifica-
 151 tion of the outlier subset $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$ as the following optimization problem:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

152 Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted
 153 label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$.
 154 The above optimization problem aims to identify
 155 a subset $\mathcal{S}$ in $\mathcal{S}_{\text{wild}}$ that minimizes the distance be-
 156 tween the EWM of the gradients and the average
 157 gradient $\bar{\nabla}_{\text{in}}$. According to Figure 1, such a subset,
 158 denoted by $\mathcal{S}_{\text{in}}^*$, may well represent the InD data in
 159 $\mathcal{S}_{\text{wild}}$, in which case $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ may capture
 160 the OOD data in $\mathcal{S}_{\text{wild}}$.

161 Solving the optimization problem in (4) can be
 162 computationally prohibitive, especially as the size
 163 of $\mathcal{S}_{\text{wild}}$ increases. To address this, we propose a
 164 greedy approximation based on a leave-one-out ap-
 165 proach, as outlined in Algorithm 1. The algorithm
 166 implements an iterative procedure for outlier detec-
 167 tion from a wild dataset, leveraging deviations in gradient information relative to the InD dataset.

168 The algorithm begins by computing the EWM of the wild data gradients at each iteration, which
 169 serves as a reference for comparing against the average gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote by d_t ,
 170 the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data left in the
 171 wild set, represented by $\mathcal{S}$, i.e., $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively designates
 172 as OOD those samples in $\mathcal{S}$ whose removal produces the largest decrease in the L_2 distance relative
 173 to d_t . Concretely, for the remaining sample set $\mathcal{S}$, each sample $i \in \mathcal{S}$ is temporarily removed, after
 174 which the EWM $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and its distance from $\bar{\nabla}_{\text{in}}$, namely $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$, are

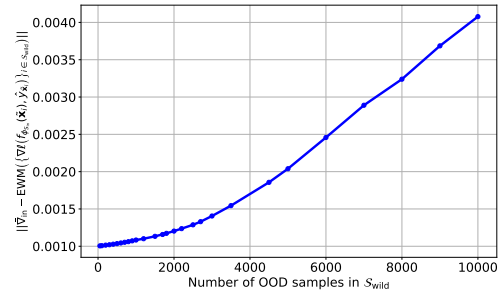


Figure 1: Distance deviation as we increase OOD samples in $\mathcal{S}_{\text{wild}}$.

175 computed. The reduction $\delta_i = d_t - \|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ is then used to select the k samples
 176 with the largest drops, which are identified as OOD. Iteration continues until δ_i no longer exhibits a
 177 significant decrease or until the maximum number of iterations is reached. The convergence criterion
 178 requires the inter-iteration change in L_2 distance to fall below a predefined threshold ϵ ; this rule is
 179 motivated by the monotonic trend in Figure 1, since, after a substantial portion of OOD samples has
 180 been removed, the L_2 distance should decrease toward a small value, indicating that further removal
 181 risks erroneously labeling InD samples as outliers and thereby degrading performance.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for** each $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

182 3.2 Training the OOD Detector with Candidate Outliers

183 After extracting the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we train the OOD detector g_θ
 184 according to the protocol of Du et al. (2024a), with the objective of sharpening the distinction between
 185 labeled InD data and candidate outliers. The objective explicitly imposes separability at the decision
 186 boundary, thresholded at 0, by assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative
 187 outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The loss is therefore defined as follows:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

188 Because the 0/1 loss is non-differentiable, we employ a binary loss induced by a differentiable sigmoid
 189 function as a smooth surrogate, preserving alignment with the original objective while rendering
 190 optimization tractable. The learning framework also incorporates the InD risk from Equation 2,
 191 thereby protecting predictive accuracy on InD samples. This unified optimization procedure improves
 192 the generalization of g_θ and supports robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as the
 193 empirical results confirm.

194 4 Theoretical Analysis

195 We next present theoretical guarantees for the filtering stage of Medix. The following theorems
 196 establish high-probability upper bounds on the misclassification rates for both InD and OOD points,
 197 thereby formalizing the two-sided robustness of EWM filtering. Detailed proofs are deferred to
 198 Appendix C.

199 **Theorem 4.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points*

in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

200

201 Theorem 4.1 shows that the inlier misclassification rate is determined by two principal components: a
 202 *concentration term*, which diminishes as the sample size grows and captures finite-sample fluctuations
 203 in the InD gradients, and a *contamination term*, which measures the extent to which OOD points
 204 can influence the EWM. Importantly, the bound remains controlled whenever the contamination
 205 ratio satisfies $\pi < 0.5$, highlighting the resilience of median-based filtering under moderate OOD
 206 contamination. We therefore turn to the complementary question: *how many true outliers fail to be*
 207 *flagged and are mistakenly retained?* The next theorem addresses this issue by bounding the OOD
 208 misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

209 Theorem 4.2 complements the preceding result by controlling the fraction of OOD points that are
 210 mistakenly retained as InD. Taken together, Theorems 4.1 and 4.2 provide a two-sided robustness
 211 guarantee for the Medix filtering procedure. The resulting inlier and outlier misclassification rates
 212 reflect an interaction among three effects:

213 **Contamination Effect.** In Theorem 4.1, the median-based criterion remains stable even when
 214 the wild dataset contains a nontrivial fraction π of OOD points, provided $\pi < 0.5$. This stability
 215 appears through the $\pi/[2(1 - \pi)]$ term, which grows with π and encodes the risk that OOD gradients
 216 distort the EWM. Conversely, Theorem 4.2 contains the symmetric penalty $\frac{1-\pi}{2\pi}$, which captures the
 217 additional difficulty of isolating OOD points when they are underrepresented.

218 **Concentration Effect.** Both bounds leverage the sub-Gaussian behavior of gradient coordinates.
 219 For InD data, this behavior implies that most gradients concentrate around $\bar{\nabla}_{\text{in}}$, preserving median
 220 stability even in finite samples. For OOD points, concentration around μ_{out} makes it possible to
 221 quantify the probability that an atypically aligned OOD sample evades the filter. These concentration
 222 contributions decay at the rate $1/\sqrt{m}$, where m denotes the relevant class-specific sample size.

223 **Separation Effect.** Appearing only in Theorem 4.2, this effect specifies how far the OOD mean
 224 gradient must be from the InD mean gradient for reliable OOD rejection. The exponential term
 225 $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ encodes the associated trade-off: greater distributional separation makes it
 226 correspondingly less likely that OOD gradients will deceive the filter.

Theorems 4.1 and 4.2 jointly imply that, with high probability, median filtering can robustly separate InD and OOD samples in unlabeled mixtures. The fraction of InD samples misclassified as outliers is controlled by contamination and concentration effects, whereas the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results furnish rigorous theoretical assurance that Medix limits both forms of error under mild assumptions.

227

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The assumption that each coordinate of the InD-sample gradients is sub-Gaussian is central to the derivation of concentration bounds and to the resulting robustness of the statistical estimation procedure. The empirical evidence in Figure 4a supports this assumption: the histogram of InD gradient values is approximately bell-shaped and concentrated around the mean, indicating light-tailed behavior, which is the defining property of sub-Gaussian random variables and implies that extreme gradient values are rare. In addition, Figure 4b compares the empirical quantiles of the gradient distribution with the theoretical quantiles of a Gaussian distribution in a Q-Q plot (Wilk and Gnanadesikan, 1968). The close alignment with the 45-degree reference line indicates that the empirical gradient distribution closely resembles a Gaussian distribution; because Gaussian variables form a subset of sub-Gaussian variables, this observation further supports the assumption. *We also provide a looser version of these bounds, which removes the sub-Gaussian assumption and holds under merely bounded second moments. This version is presented in Theorem C.3 (Appendix C.3). Although the rates deteriorate, the core robustness guarantee of Medix persists in this broader setting.*

5 Experiments

This section evaluates the efficacy of Medix across a diverse collection of InD–OOD dataset pairs and benchmarks it against 20 widely used baselines. All experiments are conducted on hardware equipped with NVIDIA A100-SXM4-80GB GPUs, and the code required to reproduce the results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We adopt the experimental protocol of Katz-Samuels et al. (2022a), which introduced the task of learning OOD detectors from wild data and thereby enables a fair comparison with prior work. Specifically, CIFAR-10 and CIFAR-100 serve as the InD datasets ($\mathbb{P}_{\text{in}}$), while OOD evaluation uses a suite of natural-image datasets, namely PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015), as OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) using the default mixing parameter $\pi = 0.5$. For instance, when PLACES365 is used as the OOD test set, the wild mixture is constructed by combining CIFAR with PLACES365, and PLACES365 is then used for OOD testing; the same procedure is repeated across all OOD datasets and baselines. The InD CIFAR dataset is divided into two halves: the first 25,000 images are used to train $\phi_{\mathcal{S}_{\text{in}}}$, whereas the remaining images are used to generate the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation, we use the penultimate-layer weights, which have been shown to be especially informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. Evaluation uses three standard metrics: (1) the False Positive Rate of OOD samples when the True Positive Rate of InD samples is 95% (FPR95 $\downarrow$), (2) the Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD classification accuracy (InD Acc $\uparrow$).

Baselines. Our comparison includes a broad set of competitive baselines, organized according to whether they are trained using only InD data or using both InD and wild data. For methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix with established OOD detection methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022); these approaches rely on a model trained with softmax cross-entropy loss. To broaden the comparison, we also include contrastive-loss-based methods such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). For methods that exploit both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by encouraging lower confidence or higher energy on outlier data. We further include WOODS (Katz-Samuels et al., 2022a), which introduced wild unlabeled data for OOD detection through constrained optimization, as well as more recent baselines including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), in order to provide a more comprehensive evaluation.

5.2 Experimental Setup

Consistent with WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is instantiated as a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and width factor 2. It is trained with stochastic gradient descent using momentum 0.9, weight decay 0.0005, and initial learning rate 0.1; optimization proceeds for 100 epochs with cosine learning-rate decay, batch size 128, and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, with dataset size and OOD performance used to guide selection. For the OOD classifier g_{θ} , we initialize from the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and append a linear binary-classification layer operating on penultimate-layer features. This classifier uses learning rate 0.001 and is fine-tuned for 100 epochs following Equation (5). The binary classification loss is combined with the InD classification loss, with weight 10 assigned to the binary component, while all remaining training parameters match those used for $\phi_{\mathcal{S}_{\text{in}}}$.

Table 1: OOD detection performance of Medix and competing baselines with CIFAR-10 as the InD dataset. Results are averaged over five runs, and the best entries are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average			
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		
	Using $\mathbb{P}_{\text{in}}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84	
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84	
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84	
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84	
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84	
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84	
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84	
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84	
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17	
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19	
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12	
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24	
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74	
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58	
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64	

Table 2: OOD detection performance of Medix and competing baselines with CIFAR-100 as the InD dataset. Results are averaged over five runs, and the best entries are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average			
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		
	Using $\mathbb{P}_{\text{in}}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96	
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96	
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96	
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96	
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96	
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96	
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96	
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96	
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90	
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78	
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38	
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76	
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91	
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33	
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83	

5.3 Results

We present our main results in Table 2 on CIFAR-100, where Medix remarkably outperforms all OOD detection baselines. The results highlight the following key observations: (1) Methods trained on both InD and wild data significantly outperform those trained exclusively on InD data. Medix reduces the FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES compared to KNN+, demonstrating the effectiveness of incorporating in-the-wild data for model regularization. (2) Medix further outperforms competitive methods utilizing wild data ($\mathbb{P}_{\text{wild}}$): On CIFAR-100, Medix achieves an average FPR95 of 5.42%, which represents a 1.32% improvement over WOODS. Additionally, Medix maintains a competitive InD accuracy of 73.33%. This slight difference can be attributed to the fact that our method is trained on 25,000 labeled InD samples, while baseline methods, which do not leverage wild data, use the full CIFAR-100 training set of 50,000 samples. We present the results for CIFAR-10 in Table 1, where Medix surpasses all baseline methods. Medix outperforms

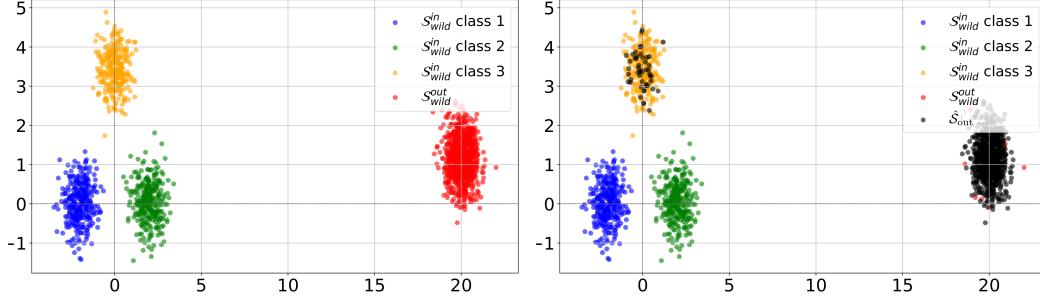


Figure 2: Illustrative application of Medix to unlabeled wild data. (a) Configuration of wild InD data S_{wild}^{in} and wild OOD data S_{wild}^{out} , where inliers are sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{S}_{out}$ selected by Medix (black), for which the error rate due to included InD data S_{wild}^{in} is only 12.5%.

WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in terms of FPR95, demonstrating its effectiveness in detecting OOD samples.

A representative visual example of Medix. We further examine the outlier-extraction behavior of Algorithm 1 on wild data S_{wild} through a two-dimensional synthetic visualization. The simulation is intentionally simple so that the filtering behavior can be inspected directly. InD data are generated from three multivariate Gaussian distributions corresponding to three classes, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, and with covariance matrix $0.25 \cdot I$ for all classes. OOD data are drawn from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, the method identifies outliers whose majority agrees with the ground truth, and Medix flags 87.5% of actual OOD samples as outliers, underscoring its robustness in outlier extraction.

Additional studies and insights. Because of space constraints, additional experiments and diagnostic analyses are deferred to Appendix A. These include: (1) ablations on hyperparameter selection and sensitivity (Appendix A.2), which show that Medix is strongly robust to hyperparameter variation; (2) a comparison between EWM and the geometric median (Appendix A.1), indicating that EWM is more sensitive to distributional shifts and therefore a more effective and reliable filtering choice in our method; (3) a comprehensive comparison with methods using competitive contrastive learning objectives (Appendix A.3), showing that Medix outperforms even these strong baselines by a substantial margin; (4) evaluation on large-scale data under the complex unseen-OOD setting, where $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), again showing that Medix substantially outperforms baselines; (5) computation and memory profiling for Medix (Appendix A.6); (6) evaluation of pseudo-label quality (Appendix A.5), demonstrating resilience to noisy or low-confidence labels; and (7) comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has attracted substantial recent attention (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One prominent line of work detects OOD data through scoring functions that characterize distributional discrepancy, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A second line of work incorporates OOD-oriented regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020;

Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). Such regularizers may be designed either to reduce model confidence on OOD data (Lee et al., 2017; Hendrycks et al., 2019) or to increase the associated energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022). Most methods in this family, however, presuppose access to an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the restriction to labeled data by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), although they do not provide a clear mechanism for explicit outlier detection. By contrast, Du et al. (2024a,b) introduce explicit outlier filtering, but their thresholding strategy is fundamentally distinct from our median-centric procedure. In addition, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch contains a prescribed ratio of InD and OOD samples; for large outsourced datasets, however, this kind of structured mixing is typically unavailable because data are randomly intermixed at the dataset level. Medix addresses this limitation by operating under dataset-level mixing without requiring batch-level structure. A related body of work studies positive-unlabeled learning, in which classifiers are trained from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). The key distinction is that these methods focus on differentiating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ without simultaneously training an OOD classifier, whereas Medix introduces a median-centric outlier-identification mechanism with provably low error rates.

7 Conclusions

This work has developed Medix, a median-centric framework for OOD detection that converts unlabeled in-the-wild data from an ambiguous deployment artifact into a statistically useful source of candidate outliers. By exploiting the intrinsic robustness of the median operation, Medix filters outliers from mixed unlabeled data and, through their subsequent combination with labeled InD observations, enables the training of a reliable OOD detector without requiring labeled OOD supervision. The theoretical analysis establishes provable bounds on the inlier misclassification rate, showing that Medix remains robust even under substantial OOD contamination (up to 50%), with the resulting errors governed by sub-Gaussian concentration and contamination effects; it also supplies complementary bounds on OOD misclassification under clear separation conditions, thereby characterizing both sides of the filtering error. Empirical validation across diverse benchmarks demonstrates Medix’s superiority over 20 baselines: relative to strong methods such as KNN+, it reduces average FPR95 by 40.98%, outperforms state-of-the-art approaches including WOODS and DRL, and attains an outlier-extraction error rate as low as 12.5%. Taken together, these theoretical and empirical results indicate that median-based filtering provides a principled and practically effective mechanism for leveraging unlabeled wild data in open-world OOD detection.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural methodological question is why Medix uses the element-wise median rather than the geometric median (Acharya et al., 2024) for outlier detection. To address this question, we conduct a preliminary experiment with CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ contains 5k CIFAR-10 samples that are disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$, which is used to compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild-dataset gradients and the proportion of OOD samples removed. As Figure 3 shows, for the same number of OOD samples in the wild dataset, the element-wise median identifies a substantially larger fraction of OOD samples as outliers than the geometric median. This behavior indicates that the element-wise median is more sensitive to distributional shifts, making it a more effective and reliable filtering choice for Medix.

Table 3: Sensitivity of OOD detection performance to the Medix hyperparameters ϵ and k .

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We perform an ablation study to assess the sensitivity of Medix to the hyperparameters ϵ and k , using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As Table 3 indicates, Medix is strongly robust to variation in both ϵ and k . In particular, Medix attains an FPR95 of 0.16 with $\epsilon = 0.005$ and $k = 20000$. Even when these values are perturbed, for example by reducing ϵ to 0.0005 or halving k to 10000, performance remains competitive, with FPR95 values of 0.20 and 0.68, respectively, reflecting graceful degradation rather than abrupt failure. These results suggest that although careful hyperparameter selection is beneficial, Medix remains effective across a range of choices and continues to outperform the baselines. This relative insensitivity to exact tuning is advantageous in real-world deployments, where exhaustive hyperparameter search may be infeasible.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to include competitive baselines with diverse contrastive learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although Medix does not use a contrastive objective, these methods are included to ensure a comprehensive evaluation. We also incorporate more recent methods, including CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). In this experiment, CIFAR-100 is used as the InD dataset and a ResNet-34 model¹ is trained following

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

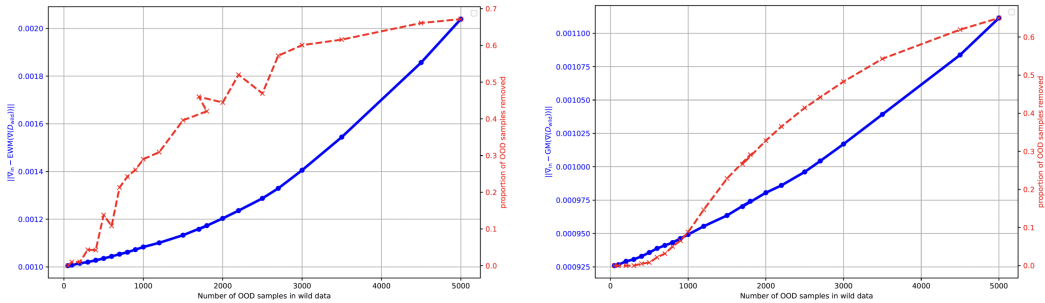


Figure 3: Comparison of the element-wise median (EWM) and the geometric median (GM) as distribution-shift-sensitive filtering statistics.

Table 4: OOD detection performance of Medix and recent competitive baselines with CIFAR-100 as the InD dataset. Results are averaged over five runs, and the best entries are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

the setup in Ming et al. (2023); Zhang et al. (2024). The model is optimized by stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, using initial learning rate 0.5 with cosine scheduling and batch size 512. We use the checkpoints provided by Ming et al. (2023)². As Table 4 shows, Medix outperforms these competitive baselines by a substantial margin, including methods trained with contrastive learning objectives, achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Moreover, Medix supplies provably low error rates, offering theoretical guarantees that most alternatives lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next examine whether Medix remains effective with large-scale wild OOD data in the unseen-OOD setting. This regime is challenging for two reasons: (1) it uses a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), thereby increasing both data complexity and computational burden; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately chosen to differ from the wild OOD distribution used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which more closely reflects difficult deployment conditions. We use CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data. The setup is especially complex because the large-scale wild OOD data (300K Random Images) differs substantially from the OOD test data (SVHN). We compare Medix with baselines that also use wild data during training. The results in Table 5 show that Medix outperforms these baselines, achieving substantially lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6), thereby demonstrating its ability to distinguish InD and OOD data under distributional mismatch.

Table 5: OOD detection performance in the large-scale unseen-OOD setting, where the wild OOD source and test OOD distribution differ.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

Another important question is whether pseudo-label quality, represented by $\hat{y}_{\tilde{x}_i}$ and especially by softmax confidence, affects OOD filtering, and whether a simple pre-filter that removes low-confidence pseudo-labels could improve model robustness. To investigate this issue, we evaluate the effect of filtering low-confidence pseudo-labels on OOD filtering using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Specifically, we compute softmax probabilities for each pseudo-labeled sample and discard low-confidence predictions using threshold 0.6; this procedure removes 15.98% of samples from the training set.

²<https://github.com>

The results show virtually no performance difference between the filtered and unfiltered variants. Both obtain FPR95 of 0.01%, while filtering increases AUROC only marginally, from 99.98% to 99.99%. Consequently, removing low-confidence pseudo-labels does not materially affect the model’s robustness or OOD detection ability, indicating that Medix is resilient to noisy or low-confidence labels and that additional filtering is unlikely to produce meaningful gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next assess whether the Medix filtering phase remains computationally feasible and memory efficient when applied to larger unlabeled collections. To this end, we conduct profiling experiments on an NVIDIA A100-SXM4-80GB GPU using approximately 15,000 unlabeled samples for each InD–OOD pair, with CIFAR-10 and CIFAR-100 serving as InD datasets and LSUN-Resize serving as the OOD dataset. As Table 6 shows, GPU memory usage remains modest, and filtering completes within a tractable time horizon even when thousands of unlabeled samples must be processed. These measurements indicate that Medix filtering is computationally practical at the examined scale and does not impose excessive memory overhead on large datasets.

Table 6: Computation and memory profile of the Medix filtering stage on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

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A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several substantive respects. For example, OpenML Saito et al. (2021) addresses outlier handling in semi-supervised learning (SSL) while training a standard classifier, whereas our setting centers on detecting OOD samples from unlabeled wild data and training a dedicated OOD detector. Thus, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) seek to improve classification in the presence of outliers, Medix improves OOD detection in an open-world setting where labeled OOD data are unavailable.

Second, SSL methods typically treat OOD samples as noise and train a classifier that remains robust despite their presence, whereas our objective is to explicitly identify OOD samples and use them to train an OOD detector. In other words, SSL methods suppress the influence of OOD samples to improve SSL performance, while Medix actively detects and leverages those samples to construct a more effective and reliable OOD detection system.

Finally, the technical mechanisms are distinct. OpenMatch relies on consistency regularization, enforcing agreement across stochastic transformations of the same input through soft constraints on classifier outputs and thereby encouraging smooth decision boundaries. Medix, by contrast, uses median-based filtering for outlier detection, exploiting the statistical robustness of the median and its insensitivity to extreme values to identify candidate OOD samples. The accompanying theory derives error bounds from contamination effects, determined by the OOD proportion, and concentration effects, induced by the sub-Gaussian behavior of InD gradients, to ensure that OOD detection error remains low.

B Broader Impacts and Limitations

As machine learning systems are increasingly embedded in consequential decision-making environments, the capacity to recognize distributional novelty becomes a prerequisite for robust, reliable, and accountable deployment. Medix contributes to this objective by providing a median-centric framework that strengthens OOD handling and, consequently, supports safer model behavior when test-time inputs depart from the training distribution. Its potential significance is therefore not confined to theoretical refinement: in domains such as healthcare, autonomous systems, and finance,

748 more dependable OOD detection can reduce deployment risk, limit unwarranted model confidence,
749 and promote greater institutional trust in machine learning technologies. Future work could pro-
750 ductively examine integration with generative models for outlier synthesis, as well as extensions to
751 dynamic environments in which OOD distributions evolve over time rather than remaining fixed after
752 deployment.

753 A potential limitation of Medix is its reliance on a mixture of unlabeled InD and OOD data, which,
754 in practical collection pipelines, may be noisy, corrupted, temporally biased, or inconsistently
755 sampled. This assumption, however, is not idiosyncratic to the proposed method; rather, it reflects
756 the uncertainty inherent in real deployment environments and constitutes a standard starting point in
757 robust learning theory. To render the assumptions explicit and assessable, we formalize them in the
758 main theorems and provide a geometric condition in Remark 4.3 that justifies the separation required
759 for reliable filtering. We also extend the guarantees to weaker settings in Theorem C.3, thereby
760 removing the need for sub-Gaussian tails and demonstrating resilience under looser distributional
761 assumptions. Thus, although Medix necessarily relies on simplifying assumptions, those assumptions
762 are interpretable, partially relaxable, and analytically connected to observable filtering behavior,
763 providing a principled foundation for future work on OOD detection under realistic noise and
764 sampling conditions.

765 C Proof of Theorem

766 To maintain clarity and notational consistency across the theoretical and empirical sections, Table 7
 767 summarizes the principal notation used throughout the paper. The table defines gradient-related
 768 quantities, sample sizes, filtering terms, and the parameters appearing in the concentration bounds.
 769 Unless otherwise specified, boldface denotes vectors, $\mathcal{X}$ denotes the input space, and all gradients are
 770 computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary for the theoretical analysis; bold symbols denote vectors and $\mathcal{X}$ denotes the input space.

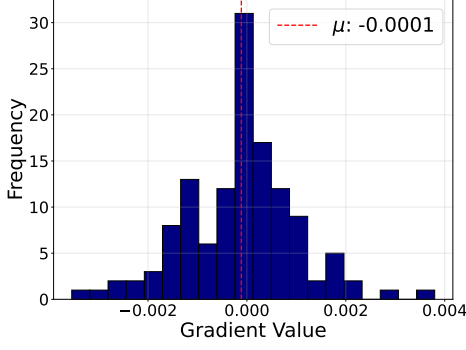
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

771 C.1 Bound on the Inlier Misclassification Rate

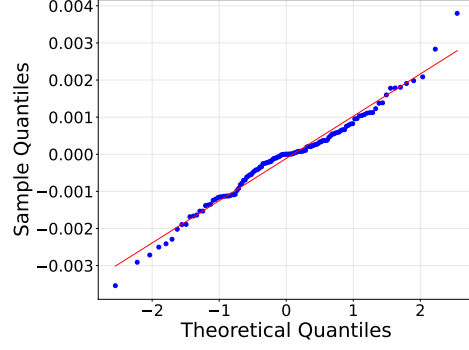
772 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 773 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 774 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 775 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

776 *Proof.* The proof proceeds in two coordinated steps. In **Step 1**, we derive a high-probability concen-
 777 tration bound showing that InD-sample gradients remain close to their mean, denoted $\bar{\nabla}_{\text{in}}$, and we call
 778 an InD sample whose gradient lies inside this high-probability region a *good inlier*. In **Step 2**, we use
 779 a “swapping” argument for the EWM objective to show that excluding more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$
 780 fraction of *good inliers*—where η denotes the fraction of InD samples outside the high-probability
 781 region—would violate optimality. Combining these ingredients yields the asserted upper bound on
 782 the inlier misclassification rate, ERR_{in} .



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

The norm comparison

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

implies that, if $\|\mathbf{v}\|_2 > \epsilon\sqrt{d}$, then at least one coordinate j must satisfy $|v_j| > \epsilon$.

Consequently, applying a union bound to (6) over $j = 1, \dots, d$ yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}. \quad (8)$$

The objective is to show that, with high probability, only a small fraction of these m_{in} inliers are bad.

Using the sub-Gaussian assumption together with the coordinate-wise union bound, define $p \in [0, 1]$ by

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

799 Linearity of expectation gives

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

800 Since X is a sum of m_{in} Bernoulli(p) variables bounded in $[0, 1]$, Hoeffding's inequality implies that,
801 for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

802 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

803 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

804 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

805 Dividing by m_{in} yields the following upper bound on the bad-inlier fraction:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

806 Denote this fraction by η . Setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ implies that, with probability at least $1 - \delta$,
807 no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients outside the
808 radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These points are the *bad inliers*; all remaining InD points are *good inliers*.

809 **Step 2: Contradiction via a Swapping Argument** We now show that if $\mathcal{S}^*$ excludes too many
810 InD points, namely if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

811 then one can construct a replacement set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of
812 $\mathcal{S}^*$, contradicting the optimality of $\mathcal{S}^*$.

813 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 814 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

815 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

816 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

817 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (26)$$

818 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 819 available for swapping in.

820 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 821 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

822 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

823 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

824 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

825 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

826 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 827 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

828 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

829 Combining this relation with the preceding inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

830 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

831 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

832 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

833 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
834 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

835 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
836 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
837 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

838 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

839 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
840 increase it, so the median distance from μ_j does not increase and possibly decreases.

841 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

842 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

843 This contradicts the assumption that $\mathcal{S}^*$ minimizes the optimization problem and therefore completes
844 the proof.

$$\Rightarrow \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

845

□

846 C.2 Bound on the Outlier Retention Rate

847 We next establish a non-asymptotic upper bound on the fraction of OOD points retained in the subset
848 returned by the MEDIX filter. The result parallels the inlier misclassification bound in Appendix C.1,
849 but it relies on separation between the OOD and InD gradient means rather than on inlier concentration
850 alone.

851 **Theorem C.2** (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point*
852 *$x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

853 **(I) Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

854 **(II) Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

855 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

856 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

857 **Proof. Step 1: Concentration of OOD Gradients.**

858 By assumption, each coordinate of $g(x)$ is sub-Gaussian and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
859 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

860 occurs with probability at most $p_{\text{out}}(\varepsilon)$.

861 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

862 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

863 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

864 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

865 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

866 which holds with probability at least $1 - \delta$.

867 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

868 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_{\varepsilon}^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

869 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_{\varepsilon} := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

870 By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of
871 entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining this bound with (47) yields the contradiction

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

Differences with the original bound. The original bound, which assumes sub-Gaussianity, features a decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

that decreases exponentially with dimension d , making the bound extremely tight for large d . In contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay term

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

that decays as $O(1/d)$, yielding a looser bound for larger d . The original bound requires sub-Gaussian tails, while the new one only assumes finite fourth moments, allowing for heavier tails. In particular, both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

that shrinks as m_{in} increases, independent of the tail assumptions.

898 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 899 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 900 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 901 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 902 bound derived from the finite fourth moment assumption.

903 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

904 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 905 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

906 and “good” otherwise. The goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 907 S_{wild} .

908 Because $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, the ℓ_2 norm can be analyzed through the sum of squared coordinates.
 909 Assume that the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$ and
 910 satisfy

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

911 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

912 The variance of v_j^2 satisfies

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

913 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

914 Applying Chebyshev’s inequality to $\|v\|_2^2$ gives

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

915 The probability of interest, $\mathbb{P}(\|v\|_2 > \epsilon \sqrt{d})$, is equivalent to $\|v\|_2^2 > \epsilon^2 d$, yielding the threshold
 916 relation

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

917 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

918 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}\left(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

919 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
920 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

921 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
922 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

923 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

924 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

925 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

926 This η upper-bounds the fraction of InD samples whose gradients deviate from ∇_{in} by more than
927 $\epsilon\sqrt{d}$ while requiring only finite fourth moments and coordinate independence.

928 **Step 2: Contradiction via a Swapping Argument** We now show that if $\mathcal{S}^*$ excludes too many
929 InD points, namely if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

930 then one can construct a replacement set $\widehat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of
 931 $\mathcal{S}^*$, contradicting the optimality of $\mathcal{S}^*$.

932 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 933 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

934 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

935 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

936 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

937 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 938 available for swapping in.

939 Construct a new set $\widehat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 940 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\widehat{\mathcal{S}}| = |\mathcal{S}^*|.$$

941 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

942 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

943 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

944 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

945 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 946 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

947 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

948 Combining this relation with the preceding inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

949 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

950 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

951 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

952 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 953 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

954 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 955 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 956 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

957 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

958 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 959 increase it, so the median distance from μ_j does not increase and possibly decreases.

960 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

961 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

962 This contradicts the assumption that $\mathcal{S}^*$ minimizes the optimization problem and therefore completes
 963 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

964

□